

Nocturnal Predictors of Morning Glucose Tolerance: The Role of Sleep and Overnight Glycemia

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Abstract - Postprandial glucose responses exhibit substantial inter-individual variability even to identical meals. While previous research has demonstrated second-meal effects and sleep disruption impacts on glucose tolerance, whether naturally occurring overnight physiological variation in free-living settings predicts next-morning glucose responses remains unclear. This study investigates whether sleep-window glycemic features and nocturnal wearable data improve prediction of standardized breakfast responses beyond meal composition and prior-evening intake.

Using the CGMacros dataset (45 participants, ~10 days each), we analyzed continuous glucose monitoring (CGM) and Fitbit data from 293 standardized breakfast events across 43 participants. We employed hierarchical mixed-effects models and leave-one-participant-out cross-validated machine learning to evaluate overnight CGM variability metrics (mean glucose, coefficient of variation) and wearable-derived features (heart rate, activity, METs), while explicitly controlling for breakfast macronutrients, prior dinner composition, dinner-to-sleep timing, and pre-breakfast glucose. The primary outcome was 2-hour incremental area under the curve (iAUC).

Counter to our hypotheses, overnight CGM and wearable features provided no incremental predictive value beyond meal-and-context baselines (likelihood ratio test $p = 0.093$ for CGM features; median leave-one-participant-out MAE unchanged at ~2,050 mg/dL·min). Within-participant decomposition revealed that when overnight glucose did associate with breakfast response, the relationship was negative—opposite the predicted direction—and largely explained by baseline-subtraction mechanics in the iAUC formula. However, an exploratory analysis revealed that prior same-participant breakfast responses substantially improved prediction (24.3% MAE reduction, $p = 2.0 \times 10^{-6}$), indicating that personalized calibration data outweighs single-night physiological snapshots.

These findings suggest that breakfast glucose responses are dominated by stable between-participant metabolic phenotypes rather than transient overnight state fluctuations. Readily available overnight wearable data from a single night does not improve next-morning glucose prediction in free-living settings but even limited personal response history does. Future personalized nutrition systems should prioritize learning individual metabolic patterns over real-time nocturnal monitoring.

Keywords - continuous glucose monitoring, postprandial response, personalized nutrition, machine learning, wearable devices, pre-registered analysis

I. INTRODUCTION

Postprandial glucose responses (which is the rise in blood glucose following a meal) typically exhibit substantial inter-individual variability even when individuals consume identical foods. This variability has important implications for metabolic health, as repeated glucose excursions are associated with increased risk of type 2 diabetes, cardiovascular disease, and other chronic conditions.

Traditionally, research has focused on meal composition as the primary determinant as primary determinant of glucose response (macronutrient content, glycaemic index, time of the food intake). However, glucose response to a meal may also depend on an individual's physiological state before eating. In particular, the overnight period, during which sleep, fasting metabolism, and circadian rhythms interact, may influence how the body responds to breakfast the next morning.

This study investigates whether nocturnal physiological state, captured through sleep-window glucose patterns and sleep behaviour, have an independent effect on next-morning glucose responses.

The overnight period represents several hours of continuous metabolic activity during which glucose regulation, sleep architecture, autonomic function, and circadian rhythms interact. Modern wearable devices (CGM and Fitbit) now enable detailed observation of these nocturnal processes outside the laboratory. However, a critical methodological challenge has limited progress: overnight glycemia is itself influenced by the prior evening meal, making it difficult to isolate whether nocturnal physiological state has an independent effect on breakfast response or whether any observed association is merely a continuation of the prior dinner's influence.

This study plans to directly addresses this challenge by investigating whether sleep-window glycaemic features and nocturnal behaviour are associated with next-morning standardized breakfast responses after explicitly controlling for dinner composition and timing

If nocturnal physiological state does influence morning glucose tolerance, it would suggest that metabolic regulation

operates on timescales longer than individual meals. It would also indicate that interventions targeting sleep quality and overnight glycaemic stability could improve metabolic health, and that current models predicting glucose responses may be incomplete by ignoring overnight data.

II. RELATED WORK

A. Second-Meal Effect Studies

The concept that one meal can influence glucose response to a subsequent meal was first demonstrated by Wolever et al. in 1991 [1]. Their seminal work showed that carbohydrate consumption at lunch could improve glucose tolerance at dinner, introducing the "second-meal effect" phenomenon. Since then, most research has focused on same-day meal interactions, particularly breakfast-to-lunch and lunch-to-dinner effects

However, the overnight period which is the longest inter-meal interval in typical eating pattern - has received considerably less attention. While dinner's immediate postprandial effects have been studied, whether overnight metabolic processes carry information about breakfast response beyond the direct influence of the prior evening meal remains unclear.

B. Sleep Disruption and Glucose Metabolism

Laboratory studies have established that sleep plays a critical role in glucose regulation. Donga et al. (2010) demonstrated that a single night of partial sleep restriction significantly impaired glucose tolerance in healthy young men. [2]

Tasali et al. (2008) showed that slow-wave sleep disruption, even without reducing total sleep time, decreased insulin sensitivity. These findings suggest that sleep quality and architecture have measurable metabolic consequences. [3]

However, these studies were conducted under controlled laboratory conditions with experimentally manipulated sleep. Whether naturally occurring variations in sleep quality and overnight glucose dynamics in free-living individuals are associated with next-morning glucose responses has not been systematically investigated.

C. CGM Based Personalized Nutrition Models

Recent large-scale studies using continuous glucose monitoring have revealed substantial inter-individual variability in postprandial glucose responses to identical meals. Zeevi et al. (2015) demonstrated that personalized dietary recommendations based on machine learning models could improve glycaemic control [4]

Berry et al. (2020) confirmed high interpersonal variability in glucose responses even among identical twins, highlighting the role of non-genetic factors [5]

D. Gap in Current Literature

The existing literature reveals three key gaps.

- First, while second-meal effects are well-documented for same-day meals, the dinner-to-breakfast carry-over has not been rigorously studied with proper controls for dinner confounding.

- Second, sleep-glucose research has been dominated by controlled interventions rather than observational studies of natural variation in free-living conditions.
- Third, personalized nutrition models have not systematically tested whether incorporating overnight CGM and wearable-derived sleep features improves prediction of morning glucose responses.

Despite the wealth of continuous glucose data collected in these studies, most predictive models focus primarily on meal composition (macronutrients, fiber, glycemic index) and a single pre-meal glucose measurement. The rich time-series data available during overnight hours remain largely unexploited in prediction frameworks. Furthermore, when overnight glucose is considered, prior studies have not adequately controlled for the confounding influence of dinner composition and timing, making it difficult to assess whether nocturnal state has an independent association with breakfast response.

III. RESEARCH QUESTION & HYPOTHESES

A. Research Question

Does sleep-window CGM-derived glycaemic features and Fitbit-derived nocturnal behaviour improve the prediction of next-morning response beyond breakfast composition, pre-breakfast glucose, prior dinner composition, and dinner-to-sleep timing?

B. Specific Hypotheses

H1 - Inferential. Overnight inter-meal glycemic features (mean Dexcom glucose and coefficient of variation, computed over the dinner-end to breakfast window) are associated with 2-hour incremental area under the glucose curve (iAUC) following a standardized breakfast shake, after controlling for breakfast composition, prior dinner and snack context, dinner-to-breakfast timing, pre-breakfast glucose, breakfast clock hour, study day, sex, age, BMI, HbA1c, and prior-day total carbohydrates and calories.

H2 - Inferential. Overnight wearable features (mean heart rate, total activity calories, mean METs over the same window) explain additional variance in 2-hour iAUC beyond the model in H1.

H3 - Predictive. Adding overnight CGM features (Block B) to a meal-and-context baseline (Block A) improves out-of-sample prediction of 2-hour iAUC under leave-one-participant-out cross-validation.

H4 - Predictive. Adding overnight wearable features (Block C) to the overnight-CGM-augmented model (Block B) further improves out-of-sample prediction.

H5 - Mechanism. The association between overnight glycemic features and next-morning iAUC, if present, reflects within-participant night-to-night variation rather than stable between-participant trait differences.

IV. METHODOLOGY

A. Data source

This study uses the publicly released **CGMacros** dataset (PhysioNet), which provides 45 participants observed under free-living conditions for approximately ten consecutive days each. The cohort comprises 15 healthy, 16 prediabetic, and 14 type-2-diabetic adults. For each participant, CGMacros provides time-aligned minute-level records containing dual continuous-glucose-monitor traces (Abbott FreeStyle Libre Pro at native 15-min resolution and Dexcom G6 Pro at native 5-min resolution), Fitbit-derived heart rate, activity calories, and METs, meal-event annotations with macronutrient composition, and meal photographs. Supplementary participant-level files contain demographic and clinical metadata including age, BMI, fasting glucose, and HbA1c.

We did not collect any new data. The contribution of this study is the analytical design described below, which reformulates CGMacros as a breakfast-event-level repeated-measures prediction problem and tests pre-registered hypotheses about overnight physiological carry-over.

B. Analytical design and unit of analysis

The unit of analysis is one **standardized breakfast shake event**. Standardized shakes were used by the original CGMacros protocol on most study mornings; this protocol-fixed breakfast provides a partially controlled morning stimulus while preserving free-living variation in everything that came before it (dinner composition, snacking, sleep, overnight physiology).

For each usable breakfast event, one analytic row was constructed containing breakfast macronutrients, prior-evening dinner and snack context, dinner-to-breakfast timing, pre-breakfast glucose baseline, overnight Dexcom-derived glycemic summaries, overnight wearable-derived summaries, and participant-level metadata.

Because each of the 45 participants contributes multiple breakfast events, the data are repeated-measures and require a model that respects within-participant clustering. We adopt a linear mixed-effects framework with a random intercept for participant (Section IV-G) for inference and leave-one-participant-out cross-validation (Section IV-H) for prediction.

C. Event selection and yield audit

Participant CSV files were scanned programmatically to identify all breakfast-labeled meal rows ($n = 434$ across the 45-participant cohort). Eligibility for each model stage required the following nested filters, each of which was applied deterministically by the pipeline rather than by manual inspection:

1. **Pre-breakfast Dexcom baseline coverage.** At least two of the three expected Dexcom readings in the 15-minute window immediately preceding breakfast must be present on the reconstructed 5-minute native grid.
2. **Post-breakfast Dexcom coverage.** At least 80% of expected readings must be present over the 0–120 min post-breakfast window, with no gap exceeding 20 min.
3. **Prior dinner match.** A valid prior dinner episode (Section IV-D) must be identifiable on the previous evening.
4. **Overnight Dexcom coverage** (Models 2 and 3). At least 80% completeness over the dinner-end-to-breakfast inter-meal window, anchored to the first native Dexcom reading at or after dinner-end.
5. **Wearable completeness** (Model 3 only). Heart rate, activity calories, and METs must each have at least 80% non-missing coverage over the overnight window. The 80% per-channel threshold (rather than full row-perfect overlap) is a pre-registered design choice intended to retain power without materially relaxing data-quality standards.

Application of these filters yielded the analytical denominators reported in

Fig. 1: 293 events across 43 participants for Models 1 and 2, and 218 events across 32 participants for Model 3. The Model 3 attrition reflects Fitbit completeness rather than glycemic data quality.

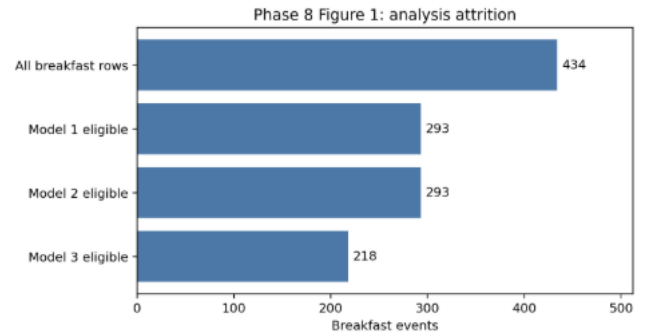


Fig. 1. Analysis attrition. Of 434 normalized standardized-breakfast events across 45 participants, 293 events from 43 participants met inclusion criteria for Models 1 and 2 (pre-breakfast Dexcom baseline coverage, $\geq 80\%$ post-breakfast Dexcom coverage, valid prior-dinner match, and $\geq 80\%$ overnight Dexcom coverage). Adding $\geq 80\%$ per-channel Fitbit completeness (heart rate, METs, activity calories) yielded 218 events from 32 participants for Model 3.

Effective sample size: Although the raw participant count is 45, the analytical sample for the primary inferential models is 293 breakfast events nested within 43 participants (~ 6.8 breakfasts per participant). The mixed-effects framework explicitly accommodates this clustering structure. For predictive evaluation, leave-one-participant-out cross-validation produces 43 (or 32 for the wearable-complete subset) held-out folds; we therefore report fold-level non-parametric tests rather than aggregate cross-validation point estimates.

D. Overnight window and prior-intake definition

The primary overnight window is defined as **dinner end to breakfast timestamp**, not a fixed midnight cutoff, because dinner timing varies meaningfully across participants and days. Importantly, CGMacros contains no sleep, wake, or bed-time annotations; we therefore use the inter-meal window as the operational overnight period throughout and avoid sleep-window terminology. Wearable channels (heart rate, METs, activity calories) over this same inter-meal window serve as proxies for overnight physiological state.

Because dinner can be split-logged across consecutive rows, dinner rows occurring within 120 minutes of one another on the same evening are merged into a single dinner episode.

Nutrient fields are summed across merged rows, and the earliest and latest constituent timestamps define dinner-start and dinner-end respectively. All snack rows occurring between dinner-end and breakfast are treated explicitly through snack count and aggregate snack carbohydrates. This treatment captures total prior-evening intake burden and reduces confounding from logging style.

E. CGM signal handling and outcome definition

Dexcom is the primary CGM source. Because the released participant files contain interpolated minute-level glucose values, variability features computed directly from the 1-min series would be biased; we therefore reconstruct glucose values on the empirically verified Dexcom 5-min native timestamp grid by timestamp filtering rather than by row-position slicing.

Libre is used only in sensitivity analyses (Section IV-I). The first 12 hours of each participant's record are dropped to remove sensor warm-up artifacts; a 24-hour warm-up sensitivity is reported in Section V-E.

The primary outcome is **2-hour incremental area under the curve (iAUC) following the standardized breakfast shake**. Concretely:

1. Breakfast timestamp defines time zero.
2. Pre-breakfast baseline is the mean Dexcom value across the 15-minute pre-breakfast window on the 5-min grid (≥ 2 of 3 expected readings; if only one is available it is used as a fallback; otherwise, the event is excluded).
3. Each post-breakfast Dexcom value over 0–120 min is baseline-subtracted; negative deviations are clipped to zero.
4. Remaining area is integrated by the trapezoidal rule.

The outcome distribution across all eligible events is shown in **Fig. 2**: median 3,118 mg/dL·min, IQR 1,561–6,268 mg/dL·min. The right skew is the motivation for our pre-specified log-iAUC sensitivity analysis (Section V-E).

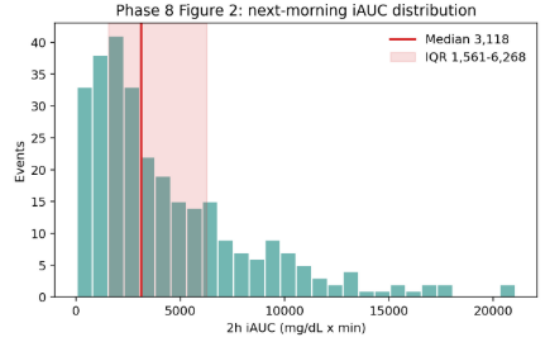


Fig. 2. Distribution of next-morning 2-hour iAUC across the 293 Model-2-eligible events

F. Feature engineering

Features are organized into three blocks of increasing data demand. The full inventory is given in **Table I**.

All features are computed by the same pipeline and cached in a single canonical event-level dataset; no model-specific feature recomputation occurs.

Table I. Feature dictionary. Block A serves as the meal-and-context baseline. Block B adds pre-registered overnight CGM features. Block C adds pre-registered overnight wearable features.

Block	Feature	Source	Formula / computation
A	Breakfast carbohydrates	Meal log	Macronutrient field, breakfast row
A	Breakfast protein	Meal log	Macronutrient field, breakfast row
A	Breakfast fat	Meal log	Macronutrient field, breakfast row
A	Breakfast fiber	Meal log	Macronutrient field, breakfast row
A	Pre-breakfast glucose	Dexcom	Mean Dexcom 5-min readings, 15-min pre-breakfast window
A	Dinner carbohydrates	Meal log	Sum across merged dinner episode
A	Dinner fat	Meal log	Sum across merged dinner episode
A	Snack carbohydrates	Meal log	Sum across snacks between dinner-end and breakfast
A	Dinner-to-breakfast interval	Meal log	Minutes from dinner-end to breakfast
A	Breakfast clock hour	Meal log	Hour-of-day of breakfast timestamp
A	Study day	Meal log	Sequential day index within participant
A	Sex, age, BMI, HbA1c	Bio	Participant-level
A	Prior-day total carbohydrates, calories	Meal log	Sum across all meals on prior calendar day
B	Overnight mean glucose	Dexcom	Mean Dexcom 5-min readings, dinner-end to breakfast

B	Overnight glucose CV	Dexcom	SD/mean $\times$ 100 over the same window
C	Overnight mean heart rate	Fitbit	Mean HR, same window
C	Overnight total activity calories	Fitbit	Sum activity calories, same window
C	Overnight mean METs	Fitbit	Mean METs/10 (raw column $\times$ 10 per data dictionary), same window

G. Statistical modeling

The primary inferential analysis fits three nested linear mixed-effects models with a random intercept for participant. Letting i index breakfast events and $p(i)$ denote the participant who contributed event i , the models are:

Model 1 (meal + context baseline):

$$\text{iAUC}_{2h,i} = \beta_0 + \beta_1 \cdot \text{breakfast_carbs}_i + \beta_2 \cdot \text{breakfast_protein}_i + \beta_3 \cdot \text{breakfast_fat}_i + \beta_4 \cdot \text{breakfast_fiber}_i + \beta_5 \cdot \text{pre_breakfast_glucose}_i + \beta_6 \cdot \text{dinner_carbs}_i + \beta_7 \cdot \text{dinner_fat}_i + \beta_8 \cdot \text{snack_carbs}_i + \beta_9 \cdot \text{dinner_to_breakfast_interval}_i + \beta_{10} \cdot \text{breakfast_clock_hour}_i + \beta_{11} \cdot \text{study_day}_i + \beta_{12} \cdot \text{sex}_i + \beta_{13} \cdot \text{age}_i + \beta_{14} \cdot \text{BMI}_i + \beta_{15} \cdot \text{HbA1c}_i + \beta_{16} \cdot \text{prior_day_carbs}_i + \beta_{17} \cdot \text{prior_day_calories}_i + u_{p(i)} + \varepsilon_i$$

$$\text{Model 2} = \text{Model 1} + \beta_{18} \cdot \text{overnight_mean_glucose}_i + \beta_{19} \cdot \text{overnight_CV}_i$$

$$\text{Model 3} = \text{Model 2} + \beta_{20} \cdot \text{overnight_mean_HR}_i + \beta_{21} \cdot \text{overnight_activity_calories}_i + \beta_{22} \cdot \text{overnight_mean_METs}_i$$

with $u_p \sim N(0, \sigma_u^2)$ and $\varepsilon_i \sim N(0, \sigma^2)$.

The random intercept u_p absorbs stable between-participant differences (chronic glycemia, basal physiology, metabolic phenotype) so that fixed-effect estimates reflect within-participant sensitivity to predictors after accounting for who the participant is. Continuous predictors are standardized.

Models are compared by likelihood ratio test (LRT) and AIC. The pre-registered support criteria require: (i) LRT $p < 0.05$, (ii) $\Delta\text{AIC} \leq -2$ in favor of the larger model, and (iii) at least one added overnight coefficient significant at $p < 0.05$ with the predicted direction. Direction was pre-specified as **positive** for both overnight mean glucose and overnight CV in Model 2, and positive for overnight mean heart rate in Model 3. Models were fit in R using lme4.

Between-participant confounders that the prof's feedback specifically flagged (age, BMI, sex, and chronic glycemia (HbA1c)) enter as fixed effects; remaining stable participant-level variance is captured by the random intercept. Time-varying confounders (dinner composition, snack carbohydrates, dinner-to-breakfast interval, prior-day intake) enter as fixed effects so that overnight coefficients are interpreted conditional on these.

H. Predictive evaluation

In parallel with inference, predictive performance is benchmarked using **leave-one-participant-out (LOPO) cross-validation**. Each fold trains on 42 (or 31) participants and tests on the held-out participant's events; this prevents participant-level leakage. Within each training fold, predictors are standardized using training-fold statistics only, then applied to the held-out participant.

Ridge regression is the pre-registered confirmatory predictive model. Ridge was selected over more flexible alternatives because (i) it is linear and therefore comparable in functional form to the LMM, (ii) it has well-controlled degrees of freedom at the available sample size, and (iii) its single regularization hyperparameter is tuned only by inner-loop cross-validation on training folds. **XGBoost** is reported as a sensitivity check.

Predictions are evaluated by per-fold MAE, RMSE, and R^2 . For the H3 (Block A $\rightarrow$ B) and H4 (Block B $\rightarrow$ C) hypotheses, the pre-registered criteria require: (i) median LOPO MAE lower for the larger block, (ii) paired Wilcoxon signed-rank test on per-fold MAE differences with $p < 0.05$ in favor of the larger block, and (iii) median $\Delta R^2 \geq 0.03$ (H3) or 0.02 (H4) with the 95% bootstrap confidence interval excluding zero. Sign convention throughout: MAE delta = larger block minus smaller block (negative improves); R^2 delta = larger block minus smaller block (positive improves).

I. Pre-registration policy and pre-specified sensitivity analyses

The full analysis plan was committed to version control under the git tag preregistration-v1 before any outcome model was fit on the eligible dataset. Pre-specified confirmatory analyses are limited to: Models 1, 2, and 3; the M1 $\rightarrow$ M2 and M2 $\rightarrow$ M3 likelihood ratio tests; ridge LOPO predictive evaluation of Blocks A, B, and C; the within-vs-between sensitivity refit of Model 2 (H5); and a mediation sensitivity removing pre-breakfast glucose. Any other analysis is reported as **exploratory**, regardless of outcome.

The pre-registered sensitivity analyses are:

1. **Log-iAUC sensitivity:** Model 2 refit with log-transformed outcome to test robustness to right-skew.
2. **Peak incremental glucose sensitivity:** secondary outcome, peak above baseline.
3. **Lagged-outcome sensitivity:** Model 2 augmented with previous same-participant breakfast iAUC.
4. **Libre sensitivity:** Model 2 refit using Libre-derived overnight features in place of Dexcom.
5. **24-hour warm-up sensitivity:** events refiltered after dropping the first 24 hours of each participant's record.
6. **HbA1c $\times$ overnight CV interaction:** heterogeneity by chronicglycemia. Three additional **post-hoc exploratory** measurement-extension analyses were

added after the primary results. These are clearly flagged as exploratory and do not revise the primary conclusion.

7. **2-hour pre-breakfast window:** overnight features recomputed over only the 2 hours immediately before breakfast. This was added to address the concern that overnight wakeups, late-night activity, or fragmented sleep might dilute a signal that lives mostly in the late-night period adjacent to breakfast.
8. **Absolute AUC outcome:** non-baseline-subtracted total AUC.
9. **Participant-centered iAUC outcome:** within-person mean removed.

V. RESULTS

A. Sample, outcome distribution, and descriptive structure

The analytical attrition pipeline (Fig. 1) yields 293 events across 43 participants for Models 1 and 2, and 218 events across 32 participants for Model 3. The next morning iAUC distribution (Fig. 2) is right-skewed with median 3,118 mg/dL·min and IQR 1,561–6,268 mg/dL·min, supporting the relevance of the pre-registered log-iAUC sensitivity analysis.

The headline finding can be summarized in one sentence: all five pre-registered hypotheses (**H1–H5**) failed their pre-specified support criteria. The four overnight carry-over hypotheses (H1–H4) returned null incremental value, and the within-vs-between mechanism hypothesis (H5) was rejected with a negative within-person association.

However, a strong post-primary exploratory finding emerged showing that within-participant breakfast-response history substantially improves prediction. The remainder of this section presents the evidence as three results: Result 1 covers H1–H4 (the carry-over null), Result 2 covers H5 (the negative within-person signal and its mechanism), and Result 3 covers the exploratory personal-history finding.

B. Result 1: Overnight CGM and wearable features did not improve inference or prediction beyond meal and context

1) Inferential models (H1, H2)

Fixed effect estimates for Models 1, 2, and 3 are shown in Fig. 3. Three patterns are immediately visible. First, breakfast carbohydrates (positive) and HbA1c (positive) are large, stable, and clearly distinguishable from zero across all three models. Second, breakfast protein, fat, and fiber show consistent negative associations with iAUC. Third, **the overnight CGM and wearable predictors that this study was designed to test (overnight mean glucose, overnight CV, overnight mean heart rate, overnight mean METs, and overnight activity calories) all have point estimates centered near zero with confidence intervals that span zero**

in every model. The forest plot makes the null visually unambiguous: the overnight block contributes no robust positive signal in the predicted direction.

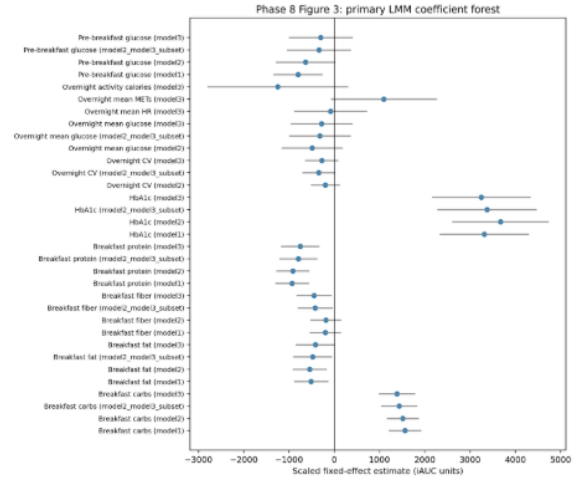


Fig. 3. Fixed-effect estimates from the primary linear mixed-effects models. Points are scaled coefficients in iAUC units; horizontal lines are 95% confidence intervals.

The nested model comparisons (Fig. 4) corroborate this. The Model 1 to Model 2 comparison is suggestive but not significant (LRT $p = 0.093$, $\Delta AIC = -0.75$; pre-registered threshold $\Delta AIC \leq -2$). The Model 2 to Model 3 comparison is clearly null (LRT $p = 0.303$, $\Delta AIC = +2.36$, i.e. the more complex model is *worse* by AIC). Neither H1 nor H2 met its pre-specified support criteria.

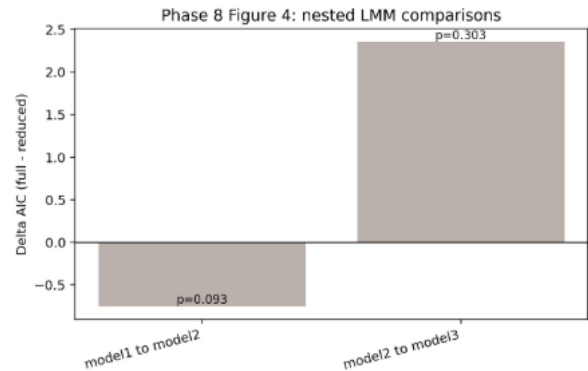


Fig. 4. Nested model comparisons. $\Delta AIC = AIC(\text{full}) - AIC(\text{reduced})$; negative values favor the larger model.

2) Predictive models (H3, H4)

The predictive analysis under strict LOPO cross-validation reaches the same conclusion. Median LOPO MAE by feature block is shown in Fig. 5. On the H3 Model-2-eligible subset (293 events, 43 folds), Block A ridge achieves median MAE = 2,047 iAUC units, equivalent to ≈ 17.1 mg/dL average error over the 120-min window, a 26.2% reduction over the naïve participant-mean baseline (median MAE = 2,772). Adding overnight CGM features in Block B does **not** improve the median: Block B ridge median MAE = 2,182, slightly worse than Block A.

The H3 paired test confirms the null: median MAE delta = -15.0 iAUC units (95% bootstrap CI: -135.6 to $+99.9$),

Wilcoxon $p = 0.483$, median $\Delta R^2 = -0.001$ (CI: -0.091 to $+0.237$). All three pre-registered criteria fail.

On the H4 Model-3-eligible wearable-complete subset (218 events, 32 folds), Block A ridge median MAE = 2,118 (17.6% reduction vs. naïve), Block B = 2,294, Block C = 2,273. The H4 paired test: median MAE delta ($B \rightarrow C$) = -41.3 (CI: -91.3 to $+16.6$), Wilcoxon $p = 0.299$, median $\Delta R^2 = +0.046$ (CI: -0.148 to $+0.126$). H4 also fails on all pre-registered criteria.

XGBoost sensitivity yielded the same conclusion (H3 sensitivity Wilcoxon $p = 0.231$; H4 sensitivity Wilcoxon $p = 0.482$).

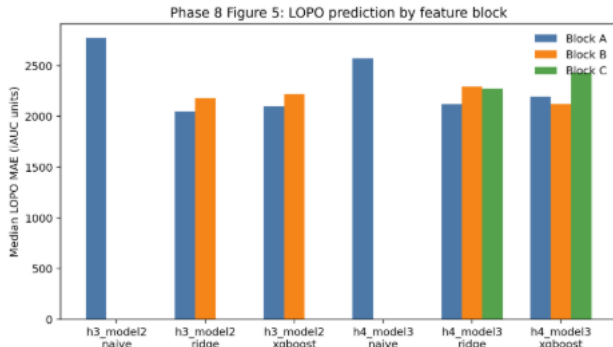


Fig. 5. Median LOPO MAE by feature block, ridge and XGBoost, on the H3 Model-2-eligible subset (left three groups) and H4 Model-3-eligible subset (right three groups).

3) Reported warning

We report transparently that on the H4 Model-3 subset, the meal-only ridge baseline (median MAE = 2,860) was worse than the naïve participant-mean baseline (median MAE = 2,789). Block A ridge nonetheless improved on naïve (median MAE = 2,576). This warning is preserved in the pipeline output and not tuned around.

4) Robustness across pre-registered sensitivities

The Phase 7 sensitivity battery (Fig. 6, Section V-E) reinforces the null direction across log-iAUC, peak incremental glucose, 24-hour warm-up, HbA1c $\times$ CV interaction, and the post-hoc 2-hour pre-breakfast window. Across these specifications the M1 to M2 LRT does not reach significance with the predicted positive overnight direction.

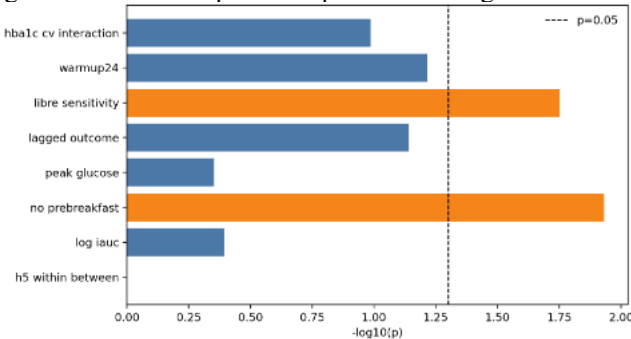


Fig. 6. Pre-registered sensitivity battery plotted as $-\log_{10}(p)$ for the relevant LRT against its reduced model. Vertical dashed line: $p = 0.05$.

The 2-hour pre-breakfast window deserves explicit mention because it directly addresses the concern that fragmented overnight sleep or wakeups could dilute a signal localized close to breakfast. Restricting the overnight CGM features to only the 2 hours immediately preceding breakfast yielded LRT $p = 0.473$, $\Delta AIC = +5.45$ which is null. The original window definition is therefore not what is hiding a true effect.

C. Result 2: When overnight glucose signal does appear, it points opposite the pre-registered direction

The within-vs-between sensitivity refit of Model 2 was pre-registered as the test of H5 (whether overnight glycemic signal reflects within-participant night-to-night variation rather than stable between-participant trait).

Decomposing each overnight predictor into a within-person deviation and a between-person participant mean produced the pattern shown in Fig. 7. The *between*-person components are positive and significant for both predictors (overnight mean glucose $\beta = +1,559$ ($p = 0.018$) and overnight CV $\beta = +943$ ($p = 0.019$)) consistent with stable trait variation.

The *within*-person components, however, point the opposite direction: overnight mean glucose $\beta = -325$ ($p = 0.019$) and overnight CV $\beta = -216$ ($p = 0.091$). H5's pre-registered support criteria required the within-person coefficient to be (i) significant at $p < 0.05$ in the predicted positive direction, and (ii) at least half the magnitude of the between-person coefficient.

The observed magnitude ratios are only 0.21 (mean glucose) and 0.23 (CV), so H5 is rejected on both the sign and magnitude criteria.

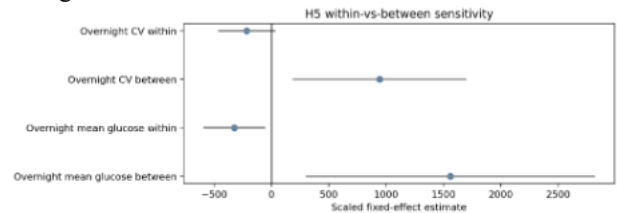


Fig. 7. H5 within-vs-between sensitivity refit of Model 2. Each overnight predictor is decomposed into a between-person participant mean and a within-person night-to-night deviation

This negative within-person association also appears in two pre-registered sensitivity specifications (Fig. 6): the no-pre-breakfast pathway model shows a significant negative coefficient on overnight mean glucose ($\beta = -829$, $p = 0.004$), and the Libre sensitivity shows a significant negative coefficient on Libre overnight mean glucose ($\beta = -901$, $p = 0.008$). Three independent specifications point the same direction with the wrong sign. We treat this as a real pattern in the data that requires explanation, not as confirmation of a positive overnight hypothesis.

Mechanism: a baseline-subtraction artifact accounts for a substantial portion

A post-primary exploratory diagnostic was run to ask whether the negative within-person effect is mathematically driven by the iAUC outcome definition. The 2-hour iAUC subtracts the pre-breakfast baseline; if higher overnight glucose elevates pre-breakfast glucose, the incremental area can mechanically shrink even when absolute glucose exposure does not.

The arithmetic check produced three quantitative results (**Fig. 8**). First, the within-person baseline-tracking slope was 0.39 (i.e. a 1 mg/dL increase in a participant's overnight mean glucose corresponds to a 0.39 mg/dL increase in their pre-breakfast glucose). Second, this baseline shift predicts a purely arithmetic iAUC reduction of -469 iAUC units per 10 mg/dL higher overnight glucose. Third, the observed raw within-person iAUC effect was -283 per 10 mg/dL. The observed/predicted ratio is **0.604** (i.e. baseline subtraction can plausibly explain about 60% of the negative within-person effect).

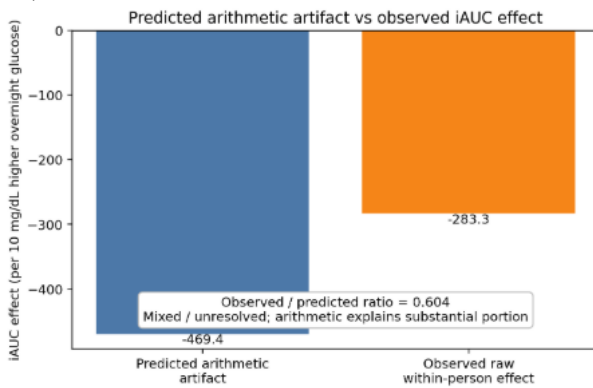


Fig. 8. Predicted baseline-subtraction artifact magnitude versus observed within-person iAUC effect. The remaining ~40% is not resolved. Comparing absolute and baseline-subtracted glucose curves stratified by high vs. low overnight glucose (**Fig. 9**, **Fig. 10**) shows the picture clearly: absolute curves are nearly identical at peak and only mildly lower in the late post-breakfast window after high-overnight nights, whereas baseline-subtracted curves diverge meaningfully because high-overnight events start from a higher baseline that the iAUC formula subtracts away. The negative within-person finding is therefore not a pure artifact but is largely consistent with baseline-subtraction mechanics. We refrain from interpreting the residual late-window divergence as biological evidence given the absence of direct mediators (insulin, sleep staging, cortisol) in CGMacros.

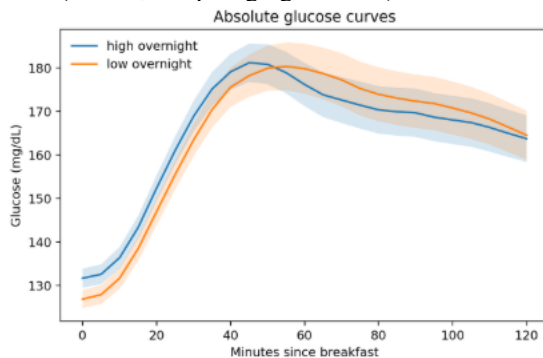


Fig. 9. Mean absolute Dexcom glucose curves over the 0–120 min post-breakfast window, stratified by high vs. low overnight mean glucose (within-person median split).

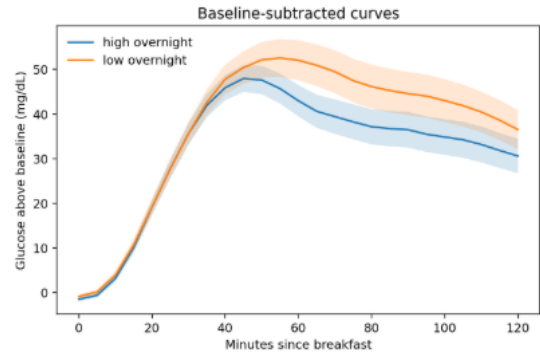


Fig. 10. Mean baseline-subtracted Dexcom glucose curves over the 0–120 min post-breakfast window, stratified by high vs. low overnight mean glucose. After baseline subtraction, the high-overnight group shows a markedly smaller incremental response across most of the window. Comparing this with Fig. 9 illustrates how the iAUC outcome definition mechanically penalizes high-baseline events even when absolute glucose exposure is similar.

The centered-iAUC exploratory model fits flagged a singular fit with near-zero participant random-intercept variance, so its strongly significant negative coefficient should not be interpreted as additional evidence for a within-person effect.

The H5 within-vs-between decomposition also produces an informative variance partition for the model overall: marginal R^2 (fixed effects only) = 0.56, conditional R^2 (fixed effects plus participant random intercept) = 0.78. The 22-percentage-point gap reflects stable participant-level differences not captured by any of the meal, context, chronic-glycemia, or overnight features. This is the variance reservoir that Result 3 (Section V-D) exploits.

D. Result 3: Within-participant breakfast-response history substantially improves prediction

A post-primary exploratory analysis (Phase 6B) tested a different question from H3/H4: not whether overnight features predict for unseen participants, but whether limited prior breakfast-response history *from the same participant* improves prediction for that participant's later breakfasts. History features were computed in a strictly forward-safe manner (only earlier events from the same participant).

On the Model-2-eligible calibration subset (250 target events across 40 participants requiring at least one prior breakfast), adding personal history to Block A reduced median MAE from 2,338 to 1,769 iAUC units, a **24.3% reduction** (Wilcoxon $p = 2.0 \times 10^{-6}$). In mg/dL-equivalent terms this is roughly 19.5 to 14.7 mg/dL average error over the 2-hour window. Even history alone (without meal context) reduced MAE to 2,026, a 13.4% reduction over Block A, indicating that prior personal response carries meaningful information independent of meal composition. Adding overnight CGM on top of personal history produced only a 1.1% additional reduction (Wilcoxon $p = 0.449$).

The qualitative improvement is visible in **Fig. 11**: in panel A on the left (strict LOPO Block A), predicted-vs-observed

scatter is heavily compressed toward the lower-left and shows weak alignment with the diagonal; in panel B (Block A + personal history), points track the diagonal substantially more closely across the full iAUC range.

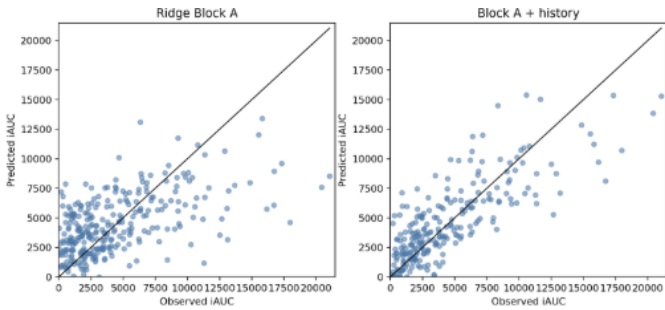


Fig. 11. Predicted versus observed iAUC under (A) strict leave-one-participant-out validation with the meal-and-context block (Phase 6 confirmatory), and (B) personalized calibration that uses prior same-participant breakfasts as additional features (Phase 6B exploratory). The diagonal is the line of perfect prediction.

This finding is corroborated within the LMM framework by the pre-registered lagged-outcome sensitivity analysis (**Fig. 12**, Section V-E). In a Model 2 augmented with previous-day breakfast iAUC for the same participant, the previous-iAUC term is positive and significant ($\beta = +580$ iAUC units, $p = 0.033$), while overnight mean glucose and overnight CV remain non-significant in the same model. The result is consistent across both modeling frameworks: prior personal response is informative, current overnight summaries are not.

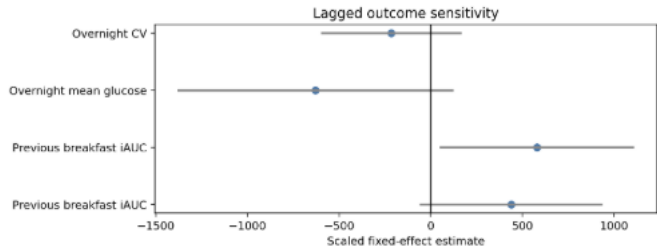


Fig. 12. Lagged-outcome sensitivity LMM coefficients. The result corroborates the Phase 6B exploratory finding (Fig. 11) within the LMM framework.

The H5 rejection (Section V-C) and the Result 3 finding fit a single coherent picture. Since predictable variation in next-morning iAUC concentrates between participants rather than within-participant night-to-night, visible directly in the 22-point gap between marginal and conditional R^2 in the H5 model, an incoming participant-specific signal (observed prior breakfasts) should be informative, whereas a within-night state signal (overnight CGM/wearable summary) should not. Both predictions are borne out.

E. Pre-registered sensitivity battery summary

The aggregated sensitivity results are shown in **Fig. 6** (axis: $-\log_{10}(p)$ for the relevant LRT vs. its reduced model). Two specifications cross the $p < 0.05$ line (Libre overnight (LRT $p = 0.018$) and the no-pre-breakfast pathway model (LRT $p = 0.012$)) but in both cases the significant overnight term carries the *opposite* sign from the pre-registered positive prediction.

We interpret these as measurement-scale and pathway sensitivity findings consistent with the negative within-person pattern of Section V-C, rather than as support for a positive overnight hypothesis. The lagged-outcome sensitivity (LRT $p = 0.072$) is informative for Result 3 (Section V-D) rather than for the overnight hypothesis. All other pre-registered sensitivities such as log-iAUC ($p = 0.404$), peak incremental glucose ($p = 0.444$), 24-hour warm-up ($p = 0.061$), HbA1c $\times$ overnight CV interaction ($p = 0.103$), are non-significant. The post-hoc 2-hour pre-breakfast window ($p = 0.473$) and absolute AUC outcome ($p = 0.169$) are also non-significant.

Across nine sensitivity specifications, no analysis supports the original positive overnight hypothesis in the predicted direction.

F. Summary of pre-registered hypothesis outcomes

Hypothesis	Test	Result	Supported?
H1	M1 $\rightarrow$ M2 LMM	LRT $p = 0.093$, $\Delta AIC = -0.75$	No
H2	M2 $\rightarrow$ M3 LMM	LRT $p = 0.303$, $\Delta AIC = +2.36$	No
H3	Block A $\rightarrow$ B ridge LOPO	MAE Δ median = -15.0 , Wilcoxon $p = 0.483$	No
H4	Block B $\rightarrow$ C ridge LOPO	MAE Δ median = -41.3 , Wilcoxon $p = 0.299$	No
H5	Within-vs-between Model 2 refit	Within- β fails both sign (negative) and magnitude (ratio 0.21–0.23 vs. ≥ 0.5 required) criteria	No

VI. DISCUSSION AND LIMITATIONS

A. Principal Findings

This pre-registered study found no evidence that overnight CGM-derived glycemic features or wearable-derived sleep proxies improve next-morning glucose prediction beyond meal composition and prior-evening intake controls. All five primary hypotheses were not supported. When overnight glucose did associate with breakfast iAUC, the relationship was negative—opposite our prediction—and largely explained by baseline-subtraction mechanics. However, prior same-participant breakfast responses improved prediction substantially (24% error reduction), indicating that stable personal metabolic phenotype dominates transient overnight state.

B. Reconciling the Null Result with Prior Literature

Our findings appear to differ from laboratory studies showing that experimentally manipulated sleep deprivation impairs glucose tolerance [12, 13], though several factors may account for this difference. First, magnitude of exposure: laboratory protocols impose severe disruptions (4-hour sleep opportunities, forced awakenings) that likely exceed the range of natural variation in our free-living cohort (median overnight

glucose CV 8.3%). Second, confounding control: laboratory studies typically compare treatment vs. control nights within-person but may not fully account for prior-meal differences across conditions; our explicit controls for dinner macronutrients, timing, and snacks represent a complementary analytical approach. Third, measurement scope: CGM captures glucose concentration but not the hormonal mediators (insulin, cortisol, incretins) that laboratory studies directly measure. Overnight glucose may reflect downstream metabolic state rather than serve as a causal driver of breakfast tolerance.

Our null result does not contradict laboratory findings; rather, it suggests that the physiological mechanisms activated by severe experimental sleep restriction may not operate detectably within the range of natural overnight variation captured by consumer wearables in free-living settings. The 2-hour pre-breakfast window sensitivity (LRT $p = 0.473$) indicates our window definition is unlikely to be masking a strong proximal effect, though subtle effects below our detection threshold remain possible.

C. The Negative Within-Person Pattern

The within-vs-between decomposition revealed a significant negative within-person association ($\beta = -325$, $p = 0.019$), replicated across three specifications. Our diagnostic analysis showed that baseline-subtraction mechanics account for ~60% of this: higher overnight glucose elevates pre-breakfast baseline, mechanically reducing iAUC even when absolute glucose trajectories are similar (Fig. 9–10). The remaining ~40% could reflect compensatory insulin secretion, glycogen depletion, or measurement noise, but without direct insulin, sleep staging, or cortisol data we cannot distinguish these mechanisms. We treat this as a methodological caution rather than evidence for a protective effect of overnight hyperglycemia.

D. Personal History Dominates Overnight State

The personal-history finding is internally consistent with the H5 variance decomposition: 22 percentage points of explained variance (conditional R^2 0.78 vs. marginal R^2 0.56) reside in stable participant-level differences not captured by any meal, chronic-glycemia, or overnight features. Breakfast responses appear dominated by trait rather than state: an individual's metabolic phenotype (genetics, insulin sensitivity, microbiome, habitual diet) predicts their response more reliably than any single night's snapshot.

Practical implication: Personalized nutrition systems should prioritize learning curves over real-time monitoring. Collecting 5–10 breakfast responses from a new user may outperform continuous overnight tracking. This extends findings from Zeevi et al. (2015) and Berry et al. (2020) by demonstrating that even recent personal data (not just cumulative training) is informative.

E. Implications for Wearable-Based Glucose Prediction

Consumer CGM systems increasingly pair glucose with wearable sleep/activity metrics. Our results suggest single-night summaries do not improve next-morning prediction when proper controls are included. Alternative use cases remain valid: long-term phenotyping (aggregating weeks of overnight data to reveal stable traits), event detection (nocturnal hypoglycemia alerts), and behavior change feedback, but not next-meal prediction.

F. Key Limitations

Sample and power: Only 43 participants for primary models (32 for wearable models); powered for medium-large effects, may miss small ones. Wide confidence intervals (Fig. 3) cannot rule out modest overnight effects.

Age heterogeneity: Participants ranged 18–69 years; metabolic and sleep physiology change substantially across this range, yet we modeled age only as a linear term. Sex hormone effects (menstrual cycle, menopause) unmeasured.

No direct sleep measurement: CGMacros lacks polysomnography, actigraphy-validated sleep staging, or sleep quality reports. Our "overnight" window (dinner-end to breakfast) includes pre-sleep wake, nocturnal awakenings, and early-morning activity. We cannot distinguish sleep from wake or measure architecture (REM, slow-wave). Wearable features (heart rate, METs) are crude proxies. However, the 2-hour pre-breakfast sensitivity also returned null, suggesting this alone doesn't explain our findings.

Nocturnal activity not fully characterized: We included summary metrics (mean METs, total activity calories) but not patterns (sustained activity bouts, timing, wake-after-sleep-onset). Fine-grained fragmentation metrics weren't derivable without additional validation.

Unmeasured confounders: Medications (antihyperglycemics, beta-blockers), stress, alcohol/caffeine, microbiome, and inflammation not captured. These likely contribute to large between-participant variance but weren't measured.

Standardized breakfast limits generalizability: Null result applies to the CGMacros shake; overnight features might interact with meal composition (e.g., predict response to high-but not low-glycemic breakfasts). Results may not extend to mixed meals or high-fat breakfasts.

Outcome definition: iAUC baseline-subtraction mechanically induces negative within-person associations (Section VI-C). Absolute AUC sensitivity also returned null, but alternative outcomes (peak glucose, time-to-peak, variability) might reveal different effects.

No mediation data: Cannot test mechanisms without insulin, C-peptide, cortisol, incretins, or metabolomics.
Generalizability: Small convenience sample, not population-representative. Findings may not extend to children, shift workers, elite athletes, pregnant individuals, or populations with different dietary/sleep cultures.

G. Methodological Contributions

Despite null primary findings, this study advances the field: (1) explicit confounding control for prior-meal effects rarely implemented in second-meal studies; (2) pre-registration discipline transparently reporting unsupported hypotheses; (3) leave-one-participant-out validation approximating real-world deployment; (4) within-vs-between decomposition isolating trait from state; (5) quantitative artifact diagnostics for derived outcomes (Fig. 8).

H. Future Directions

Key questions remain: Do long-term overnight aggregates (weeks/months of habitual patterns) predict response even though night-to-night variation doesn't? Does direct sleep staging from PSG reveal effects missed by wearables? Do overnight features interact with meal composition? What mechanistic mediators link overnight state to breakfast tolerance?

For now, our data suggest that in free-living settings, learning an individual's typical metabolic response patterns may be more informative for next-morning glucose prediction than monitoring single-night physiological snapshots.

VII. CONCLUSIONS

Overnight physiological data from consumer wearables and CGM do not improve next-morning glucose prediction beyond meal composition and prior-evening intake in free-living adults. Single-night summaries of glycemic variability and sleep proxies provided no incremental value across nested statistical models and leave-one-participant-out validation. When overnight glucose did associate with breakfast response, the relationship was negative and may be artifactual (baseline-subtraction mechanics).

In contrast, limited prior breakfast-response history from the same participant reduced prediction error by 24%, demonstrating that stable personal metabolic phenotype

dominates transient overnight state. Personalized glucose prediction systems should prioritize learning individual response patterns from calibration data over real-time overnight monitoring. While severe experimental sleep disruption clearly impairs glucose tolerance, naturally occurring overnight variation as captured by consumer devices does not carry actionable predictive information for next-morning responses when proper confounding controls are applied.

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